Zürcher Hochschule
für Angewandte Wissenschaften

Life Sciences und
Facility Management

Imaging for the Life Sciences

Master of Science in Life Sciences
CO4
Cluster Computation

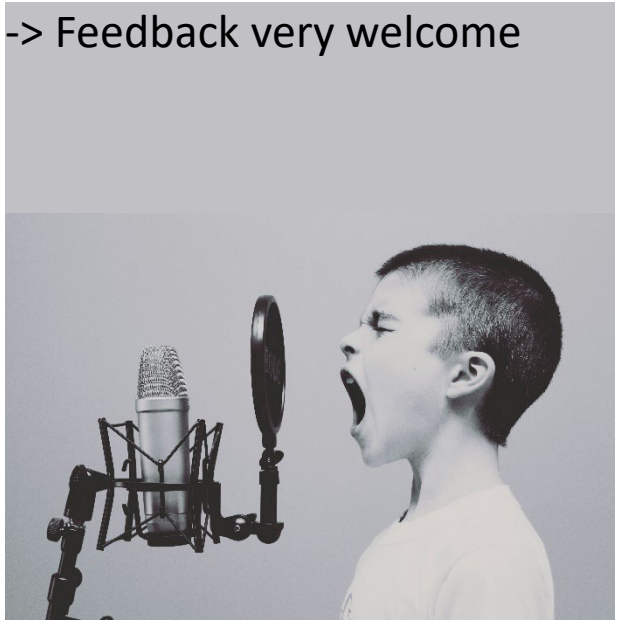
Imaging for the Life Sciences

In the module “Imaging for the Life Sciences” we will deal with:

- **Imaging techniques**
- **Image processing methods**

We will have a mixture of:

- **Theory**
- **Practical sessions**
- **Excursions**

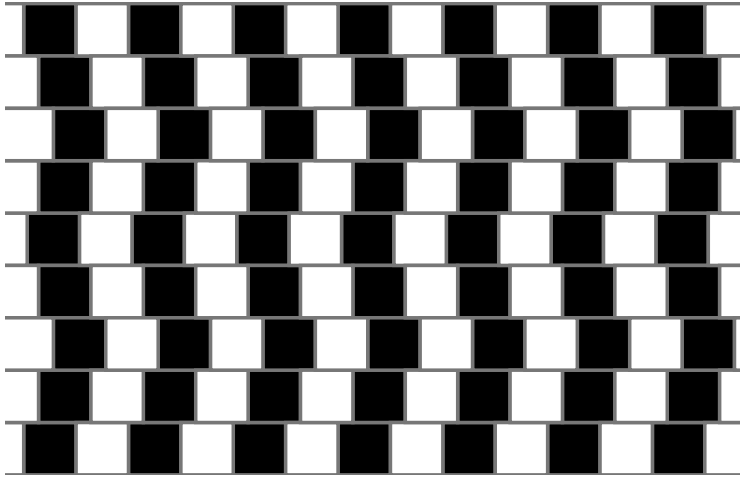


Imaging for the Life Sciences

Why we decided to design the module this way

- The human brain processes visual information faster and more intuitively than any other form of input.
- We trust images — almost automatically.

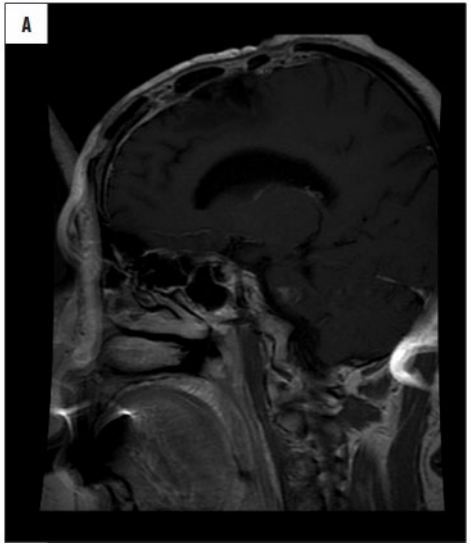
- That's exactly what makes them so powerful and dangerous.



By Fibonacci - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=1788689

Imaging for the Life Sciences

Why we decided to design the module this way



- The human brain processes visual information faster and more intuitively than any other form of input.
- We trust images — almost automatically.

- That's exactly what makes them so powerful and dangerous.
  E.g. A misleading MRI, a poorly chosen contrast, an artefact — none of these are obvious at first glance. We respond to the image, not to the reality behind it.

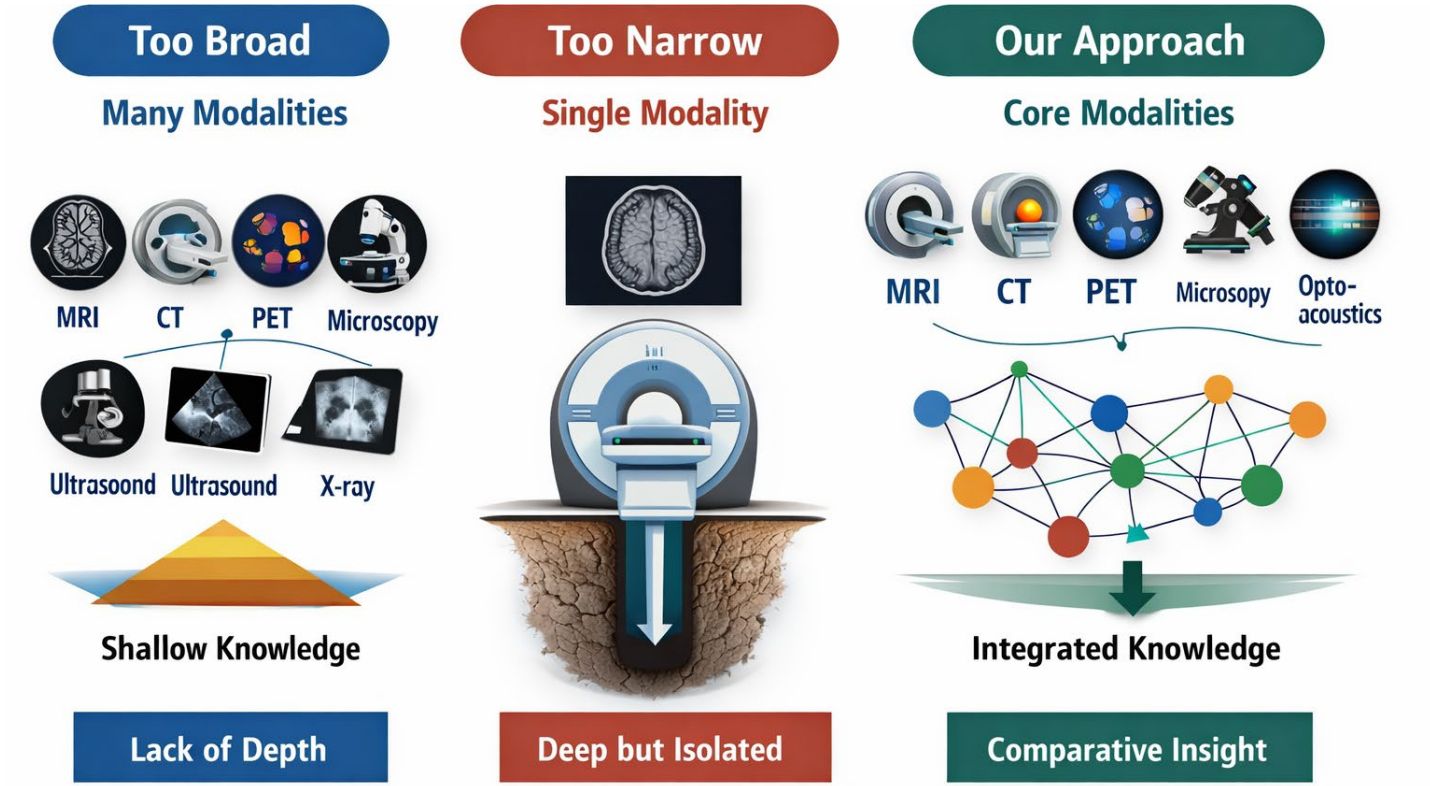
- In the life sciences, this matters. We don’t need (but like) a beautiful image. We want a correct one — an image that doesn't mislead us when making a diagnosis, analyzing a cell, or drawing a biological conclusion.

- To judge whether an image is correct (or meaningful), you need to understand how it was created.
  - What physical process produced the signal?
  - What does a filter actually change?
  - What does brightness in an MRI really mean?

- This cannot be learned by memorizing facts — it has to be worked through.

Imaging for the Life Sciences

Theoretical part: Why We Focus on Selected Imaging Modalities



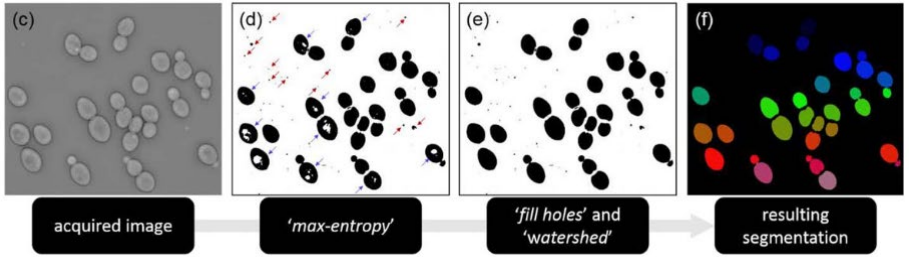
Imaging for the Life Sciences

Practical Part: From Image to Insight

What you will do in this module:

In the practical part we start with simple algorithms. Not because AI is worse — but because every step stays visible. You see what happens, you can control it, you can question it. That is the foundation for critically evaluating any tool — including AI.

- Extract features from images
- Segment biological structures
- Build simple analysis pipelines
- Focus on understanding each step
- Interpret results and connect to theory
- Goal: understanding over tools



Terenna et al., *Cytometry Part A* (2019) — PMC6585648

Imaging for the Life Sciences

Excursions

- Two exemplary high-end sites at ETH

ETH IBT: MRI research

ETH PIC

Imaging for the Life Sciences

Contributors

The lecture is based on extensive templates by our predecessors Prof. Alex Ringenbach and Prof. Dr. Antje-Christin Knopf and Prof. Dr. Markus Rudin

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Imaging for the Life Sciences CO4

16. April	OVR B125/126	FHNW, Von Roll-Strasse 10, 4600 Olten
23. April	Aula KBS	BBZ Olten, Aarauerstrasse 30, 4601 Olten / (building C)
30. April	online	
07. May	online	
14. May	Ascension day (no classes)	Self study
21. May	Morning: Excursion Afternoon: Excursion	Morning: Institute for Biomedical Engineering, ETH Zurich, Gloriastrasse 35, 8092 Zürich ETH-Zentrum Afternoon: University of Zurich and ETH Zurich, Preclinical Imaging Center (PIC), Vladimir-Prelog-Weg 1-5/10, HCI D426, 8093 Zurich, Switzerland
28. May	Aula KBS	BBZ Olten, Aarauerstrasse 30, 4601 Olten

ETH IBT: MRI research

ETH PIC

Imaging for the Life Sciences

Verification of Achievement

1. Project work (50%), about 27h
2. Written exam (closed-book, offline, 25.06.2026, 8:30) (50%)

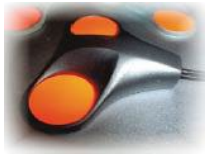
CO4

Module title	**Imaging for the Life Sciences**
Code	CO4
Degree Programme	Master of Science in Life Sciences
Group	Computation
Workload	3 ECTS (90 student working hours: 42 lessons contact = 32 h; 58 h self-study)
Module Coordinator	**Name**: Dr. Andreas Hock **Phone**: +41 (0)58 934 50 99 **Email**: andreas.hock@zhaw.ch **Address**: ZHAW Life Sciences und Facility Management, Grüental, 8820 Wädenswil
Lecturers	• Dr. Andreas Hock, ZHAW • Dr. Norman Juchler, ZHAW • Prof. Dr. Steffi Lehmann, ZHAW • Dr. Robert Vorburger, ZHAW • Dr. Luis Dean Ben, ETH Zurich
Entry requirements	• Basic knowledge of biology • Bachelor level of analysis, linear algebra, statistics and signal processing • Basic Python programming skills • An installed and functional Python programming environment. Installation instructions will be made available on the MSLS Community Centre. Students are expected to verify their setup before the first lecture. If you are unsure whether your prior knowledge is sufficient, you can test it with the quiz on this page: https://mslscommunitycentre.ch/course/view.php?id=140
Learning outcomes and competences	After completing the module, students will be able to: • Understand the techniques of different imaging modalities used in medicine and the life sciences, e.g. ultra-sound, X-rays, CT, MRI, SPECT, PET etc. • To interpret typical image data from the life sciences and (bio-)medicine • Perform basic image processing tasks, such as de-noising, morphological filtering, segmentation, either programmatically in Python or with tools like ImageJ/Fiji.
Module contents	• Imaging methods and applications to different fields in the life sciences • Image processing techniques & workflows • Student projects • Excursions
Teaching / learning methods	Lectures, accompanied with practical work
Assessment of learning outcome	1. Project work (50%) 2. Written exam (closed-book) (50%)
Format	7-weeks
Timing of the module	Spring semester, CW 16-22
Venue	Blended learning format. Presence sequences take place in Olten
Bibliography	-
Language	English

About me

Emergency Medical Technician with the German Red Cross, Esslingen, Germany

MSc Medical Engineering at Technical University of Munich



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